Jannik Wirtheim

Data Science & Machine Learning

LinkedIn GitHub Personal Website



Contact

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Schwäbisch Hall, Germany

Skills

Programming

Python, R, SQL	(intermediate)
C++, Linux	(basic)

ML Frameworks

PyTorch, OpenCV, scikit-learn, Gymnasium, SciPy, smolagents, GeoPandas

Languages

German	(native)
English	(C1)

Certifications

Hugging Face:

Deep Reinforcement Learning
Fundamentals of LLMs
Fundamentals of Agents

NASA:

Fundamentals of Remote Sensing

Research Interests

Computer Vision
Reinforcement Learning
Remote Sensing
AI for Climate

Education

M.Sc. Social and Economic Data Science (1.2) 10/2023 – 03/2026
University of Konstanz

B.Sc. Business Psychology (1.3) 10/2020 – 09/2023
Stuttgart Technical University of Applied Sciences

Bank Clerk Apprenticeship (84/100 points) 09/2016 – 02/2019
VR Bank Heilbronn - Schwäbisch Hall e.G.

Experience

Research Assistant, Data Science and Generative AI since 06/2026
Fraunhofer IAO, Heilbronn

Contributing to projects at the intersection of Data Science and Generative AI.

- Developing a system for automated picking processes using Sensor Fusion and Computer Vision in industrial parts storage systems.
- Supporting the "ExpertAIse" project to capture and preserve expert knowledge in organizations through LLM-assisted interviews.

Master's Thesis 10/2025 – 03/2026
Max Planck Institute of Animal Behavior, Konstanz

Development of a GAIL-based imitation learning framework to model multi-agent behavior in a predator-prey setting from video data.

- Developed a CV-pipeline to extract fish trajectories from video data.
- Designed and trained a GAIL framework, optimized using Evolutionary Strategies in a vectorized simulation environment.
- Evaluated the models and applied Explainable AI methods to analyze emergent behavior and collective decision-making.

Student Assistant, Computer Science Department 09/2025 – 12/2025
Diak Klinikum Landkreis Schwäbisch Hall gGmbH

Contributed to IT projects in a clinical environment and supported technical and administrative processes.

- Supported IT infrastructure and maintenance across clinical departments.
- Assisted in the design and development of a web application.

Research Internship, Social and Economic Sciences 08/2022 – 09/2023
International Performance Research Institute gGmbH, Stuttgart

Contributed to research projects focusing on statistical analysis, Data Science, and Machine Learning.

- Developed a statistical reporting tool for the research project "FlyNet".
- Designed guidelines to improve data quality for Predictive Maintenance and Machine Learning pipelines within the project "MLready".

Student Assistant, Business Psychology Laboratory 09/2021 – 08/2022
Stuttgart Technical University of Applied Sciences

Supported organizational and research activities within the laboratory.

- Provided statistical support for qualitative and quantitative studies.
- Created video tutorials for biofeedback measurement using movisens.

Bank Clerk 02/2019 – 08/2019
VR Bank Heilbronn - Schwäbisch Hall e.G.

Gained experience in banking operations with focus on client relations and advisory.

- Handled front-office tasks (cash, scheduling, complaints).
- Advised customers on financial products and supported sales.